VISTA: Situated Visual Modality Adaptation for Spoken VLM Responses in XR

Category: Research

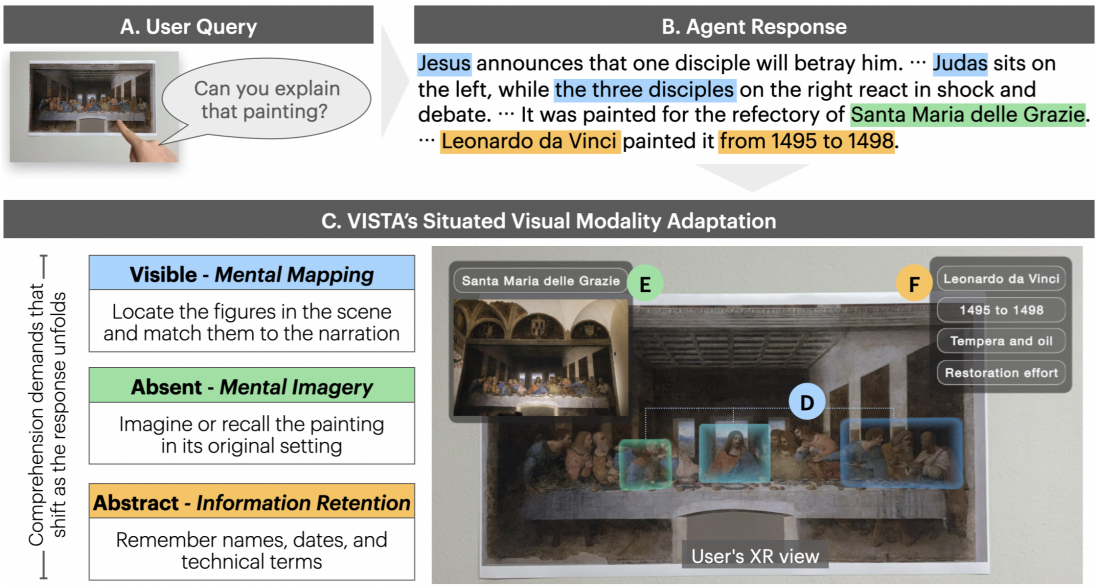


Figure 1: Example operation of VISTA. Given a user's query and egocentric view (A), the VLM generates a free-form, information-rich response (B). Situated visual modality adaptation (C) generates visual support for three comprehension demands (left) at run time. Spatial highlights (D) guide the user's gaze to each described figure, a reference image (E) shows the painting's original setting, and pinned keywords (F) keep unfamiliar names, dates, and terms in view.

ABSTRACT

We present VISTA, an XR agent system that delivers free-form, information-rich VLM responses with adaptive visual support to improve comprehension and reduce cognitive load. A response can shift among visible objects, absent content, and abstract facts, requiring users to locate, imagine, and retain information while listening. Existing XR agents, however, typically deliver every response in one fixed visual form that supports only a single demand. To address this, we present situated visual modality adaptation, an approach that (i) adapts visual support as the demand changes within a response and (ii) synchronizes each cue with the narration. The key idea is to use the response–scene relation, i.e., how the described content relates to the user's physical surroundings, to select and generate matched visual support at run time. We performed a user study and interviews with a working prototype, and participants showed improved comprehension with lower workload compared to an audio-with-text baseline.

1 INTRODUCTION

Recent advances in VLMs are transforming how XR agents help people explore their surroundings [25, 59]. Using egocentric camera streams and additional sensors, these agents interpret the user's physical context and answer a wide range of situated questions [17, 46, 77]. For example, a shopper might ask, "What is this used for?" and a museum visitor, "Tell me about this painting." Such questions elicit detailed, free-form responses, and today's XR agents handle this interaction through a voice interface, with spoken queries and audio answers [27, 56]. Consider a visitor viewing Leonardo da Vinci's *The Last Supper* (Figure 1A). The agent would respond (Figure 1B): "Jesus announces that one disciple will betray him… It was painted for the refectory of Santa Maria delle Grazie… Leonardo da Vinci painted it from 1495 to 1498." As the narration continues, the user must map each description to the correct figure (highlighted in blue), imagine a place they cannot see (green), and retain unfamiliar details (yellow). A single information-rich response can impose all of these demands at once. Delivered as a uniform audio stream, it forces the user to resolve every demand in real time, and failing any one undermines comprehension of the rest [45, 82].

Recent work has explored augmenting verbal explanations in XR with visual cues to support comprehension. Most of this work pairs a response with a single form of visual support, serving one comprehension demand. Systems built for predefined query types (e.g., "What is this?") attach anchored labels and highlights that point users to the object in question [38, 46, 72]. Agents that answer open-ended questions also keep one form for every response, whether a text transcript [27, 39, 56], an object-anchored panel [23], or embodied gestures [8, 51]. A few systems adapt their cues to the type of information, showing a highlight for a location, an arrow for a motion, and a gesture for a spatial relation [51, 73, 86]. Even with this adaptation, the support stays within a single demand, typically helping users locate and follow visible targets (Figure 2A). In contrast, free-form, information-rich responses are improvisational. Preparing visual support in advance is difficult, as the content of a response emerges only after generation [18, 21, 50] and the type of support it needs changes as the narration progresses (Figure 2B).

We present VISTA, an XR agent system that delivers free-form, information-rich VLM responses as audio narration paired with adaptive visual support to improve comprehension while reducing cognitive load. It employs situated visual modality adaptation, an approach that matches visual support to the comprehension demands of each part of a response. Understanding such a

response involves three demands: (i) mental mapping to locate each described target in the physical scene [37, 75], (ii) mental imagery to imagine or recall described content absent from the user's view [11, 62], and (iii) information retention to hold abstract details in memory as the narration moves on [45, 57]. VISTA generates the matched cues at run time and presents each one just in time with the narration, aiming to improve comprehension while reducing cognitive load.

We distinguish three response–scene relations by how the described content relates to the user's current physical environment: *Visible*, *Absent*, and *Abstract*. Returning to the Figure 1 example, our approach generates visual support on the fly and overlays it onto the user's view (C). For Visible content, spatial highlights (D) direct the user's attention to each described figure, removing the burden of searching the scene. For Absent content, a reference image (E) externalizes the original setting the user would otherwise have to imagine. For Abstract content, pinned keywords (F) hold unfamiliar names, dates, and terms in view, helping the user retain and revisit them. VISTA segments the response by these relations and provides an overview UI, so the user can grasp the scope of the response and pace its delivery.

We demonstrate VISTA across five everyday scenarios: art explanation, presentation Q&A, machine setup, product comparison, and matcha brewing. Videos of all five scenarios are available at https://vista-xr.github.io/. We evaluated each stage of the automated pipeline on 30 egocentric image–query pairs spanning six categories. We also conducted a user study with 12 users in the art explanation and beverage crafting scenarios to verify VISTA 's effectiveness compared to the audio-with-text baseline. Participants showed improved comprehension in both objective (20.3%) and self-reported measures (29.9%), and significantly lower perceived workload (22.3%) under VISTA's integrated delivery. Our contributions are as follows:

- We introduce situated visual modality adaptation, an approach that transforms free-form agent responses into adaptive visual support at run time based on how each response segment relates to the user's physical environment.
- We design and implement VISTA, an XR agent system that automatically segments free-form responses, generates matched visual support, and synchronizes each cue with the audio narration at run time.
- We demonstrate through a within-subjects user study ($N = 12$) that VISTA improves comprehension and reduces perceived workload compared with a standard audio-with-text baseline.

2 Related Work

2.1 Visual Support for XR Agent Responses

Recent XR agents use VLMs to answer a wide range of questions about the user's surroundings [33, 77, 85], and researchers have proposed visual cues that help users understand responses delivered through audio narration. Most of this work serves a single comprehension demand (Figure 2A). Systems built for predefined query types (e.g., "What is this?" or "Where did I leave my car keys?") support them with concise visual cues, such as anchored labels and highlights that point to the queried object [29, 38, 40, 46, 72]. Agents that answer open-ended questions instead use uniform visual support independent of the response content, whether a text transcript [27, 39, 56, 77], simplified text [84], an object-anchored panel [23], or embodied expressions [8]. Fewer systems adapt the visual support to the information at hand, building on a task or a set of objects known in advance [47, 73]. Guided Reality selects among highlights, arrows, and hand motions to guide each step of a machine task [86], and AgentHands generates deictic, affective (e.g., a thumbs-up), and iconic (e.g., numbers) hand gestures for pre-registered objects to support understanding [51].

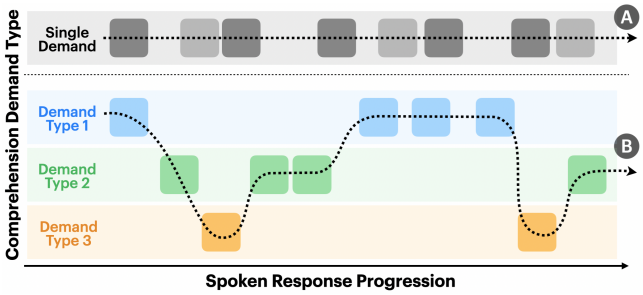


Figure 2: Prior work assumes a single comprehension demand per response (A). In a free-form, information-rich response, the dominant demand changes as the narration progresses (B), so no single form of visual support serves the whole response.

In contrast, a single free-form, information-rich response can direct users to visible objects, describe absent places or states, and mention names or values worth retaining, so the dominant demand itself shifts as the narration progresses (Figure 2B). The responses we target are improvisational. What a response will contain becomes known only after it is generated, so the visual modality must be selected across demands at run time. Our approach adapts visual support to how each described content relates to the user's physical context, addressing each demand as it arises.

2.2 In-Situ Visual Augmentation

XR has long overlaid virtual content onto physical surroundings to convey situated information [10, 28]. Prior systems have used this capability in two broad ways: (i) directing attention to relevant information already present and (ii) externalizing information that is not currently visible. The first is useful when physical surroundings are cluttered or unfamiliar: the attention funnel draws a tunnel of virtual frames from the user's view to a target [6]; highlights and bounding boxes outline or tint relevant objects and regions [36, 68, 74]; and arrows indicate locations, directions, or movements [14, 71]. These cues have supported physical tasks: classic systems overlaid assembly instructions directly on parts [76], ProcessAR authors in-situ 2D/3D instructions from expert demonstrations [16], and surgical telementoring anchors annotations in the operating field [24].

The second externalizes what the user cannot currently see, including absent objects, unfamiliar places, and future or target states. Such information has been shown as 2D reference images, 3D models, or 360° panoramas. Earlier systems prepared this content for specific sites or scenarios, from prerecorded panoramas [15] to manually authored historical reconstructions [58, 61]. Recent systems retrieve or generate it on demand: RealityTalk [48] augments live presentations with visuals triggered by the presenter's speech, RealityChat [35] retrieves reference images and maps during live conversations, and Thing2Reality [34] turns 2D content into 3D objects that participants can manipulate in XR meetings. XR thus offers a wide spectrum of visual support, from a simple box drawn over an object to a panorama that transforms the surrounding view. Making a free-form agent response easy to follow requires placing the right cue in the right place at the right moment, based on how each piece of information relates to the user's physical environment.

2.3 Audio-Visual Coordination

Humans process visual and auditory information through separate channels, so pairing visuals with audio narration can effectively deliver XR agent responses [26, 53, 60, 69]. However, simply attaching visual cues to speech without precise temporal alignment can split attention and add cognitive load.

Two factors make coordination especially critical in XR. First, the narrow field of view (FoV) limits how much content can be displayed simultaneously, so users must focus their attention deliberately [49, 84]. Second, when audio and visuals arrive out of sync, users must hold one in working memory while waiting for the other, increasing time pressure and consuming cognitive resources [2, 13, 41, 63]. AgentHands [51] shows that precisely timing gestures to speech improves agent communication, but coordinating diverse visual modalities within a single response remains underexplored.

These principles were established with authored instructional media, where the content, its visuals, and their timing are fixed in advance [54]. An XR agent's response, in contrast, is generated only at run time, and its described targets are scattered across a scene that differs with every question. Whether and how the principles carry over to such delivery remains an open question. VISTA operationalizes them at run time, aligning each visual cue with the corresponding moment in the narration so that users immediately see what is being discussed, grounding verbal descriptions in their view.

3 VISTA DESIGN

We aim to automatically transform the free-form, information-rich responses of XR agents into audio-visual presentations, with two design goals: (i) improving comprehension and (ii) reducing cognitive load. Visual support could be adapted based on response-level properties such as length, structure, or topic. These properties characterize a response as a whole and do not indicate how comprehension demands change from one part to the next. Within a single response, however, the focus continually shifts among described targets, including entities, states, and facts. Each part imposes a comprehension demand based on how its content relates to the user's physical surroundings. We capture this as the **response–scene relation** and use it as an intermediate representation to transform each part of a free-form response into appropriate visual support at run time. We distinguish three relations by the demand each imposes (Section 3.1) and design visual modalities that satisfy each relation's requirement (Section 3.2).

3.1 Response–Scene Relations

We assign a response–scene relation to each response part using two distinctions: (i) whether the described content appears in the user's current view and (ii) whether it has a useful visual counterpart. Content in view is *Visible*. Content out of view is *Absent* when it has a visual counterpart and *Abstract* when it has none. We next unpack the comprehension demand imposed by each response–scene relation and derive the corresponding requirement for visual support.

Visible. When the described target is in the user's view, the primary demand is **mental mapping**. As the agent describes a tool or region in front of them, the user must search the scene to locate that target and connect it to the narration [37, 75]. This search costs more time and attention in cluttered or unfamiliar environments [3, 74, 83], and repeated searches across targets cause the user to fall behind the narration. To ease this burden, visual support should guide the user's gaze to each target as it is mentioned.

Absent. When the described content is not present in the user's current physical environment, the primary demand is **mental imagery** [11, 62]. For example, the response could describe a tool stored in another room, the fully grown state of a visible plant, or the nighttime appearance of a place seen during the day. No search of the current scene can reveal such content, so the user must imagine or recall it while following the narration [87, 88]. Building this image occupies visuospatial working memory [70], so visual support should externalize the missing entity or state in a form that matches the description.

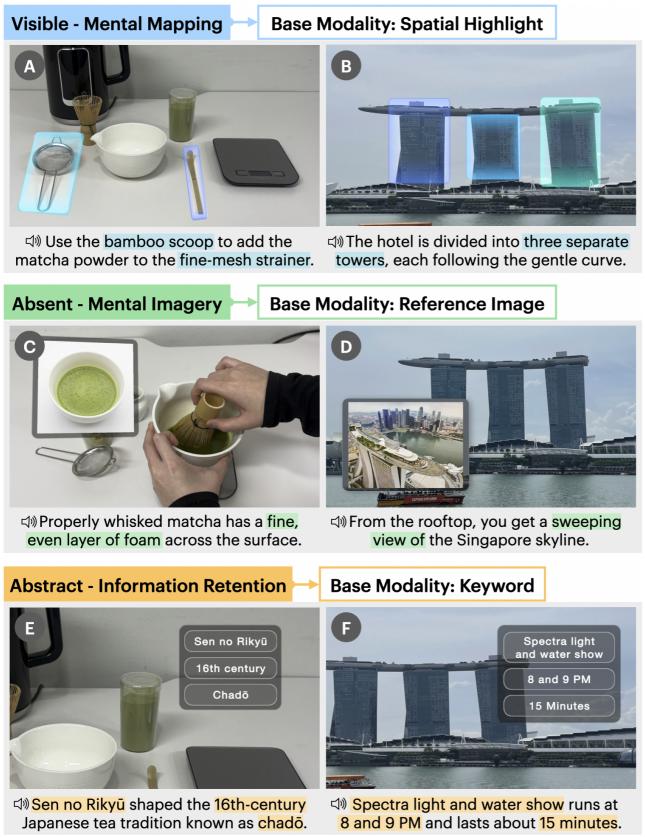


Figure 3: Examples of base modalities for visible, absent, and abstract content in matcha preparation (left) and sightseeing (right). Narration shown below each panel.

Abstract. Some information has no useful visual counterpart but remains **difficult to retain**. Unfamiliar terms, precise values, and proper names easily vanish when delivered through speech alone [45, 57]. Displaying the full response as text would keep these details accessible, but it also adds reading burden in XR [67, 81]. Visual support should instead keep only essential details in view as brief keywords that users can revisit throughout the narration.

3.2 Visual Modality Mapping

We realize each relation's requirement through visual modalities, concrete forms of visual support that reduce the cognitive load each demand imposes. For each response–scene relation, we first derive a base modality that serves its primary demand, and then introduce extended modalities for needs the base alone does not cover.

Base modalities. Figure 3 illustrates the mapping from each response–scene relation to its base modality.

- **Visible: Spatial Highlight.** We render a spatial highlight as a semi-transparent box anchored to each visible target, guiding the user's gaze without obscuring the underlying physical object (Figure 3A, B). When the response refers to multiple visible targets, each highlight receives a distinct color so the user can distinguish the targets as the narration moves among them.
- **Absent: Reference Image.** We externalize the absent content with a reference image. When image search can provide a suitable reference, we retrieve one, such as the view from the building's rooftop (Figure 3D). When the described state is specific to an object in the user's scene, a retrieved image would depict a different instance, so we generate an image that renders the state

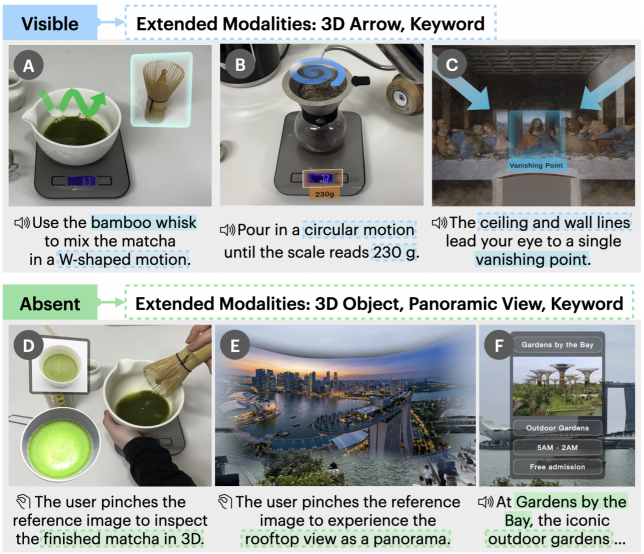


Figure 4: Extended modalities for Visible content (A–C) and Absent content (D–F).

on the user's object, such as the finished texture of the matcha in their own bowl (Figure 3C).

- **Abstract: Keyword.** We extract retention-critical details, such as exact terms, dates, values, and proper nouns, and pin them in the user's view as concise keywords (Figure 3E, F). The keywords remain visible throughout the narration, allowing the user to revisit them after they are mentioned.

Extended modalities. Base modalities address the primary demand of each response–scene relation. Some content also describes a motion to follow, a place to experience, a form to inspect, or details to retain. We fulfill these additional needs with extended modalities that augment the base. Figure 4 shows the extended modalities for visible (top) and absent (bottom) content.

- **3D Arrow for Directionality.** A spatial highlight marks where a target is but cannot express how it should move [65], leaving the user to infer the trajectory from speech alone. When the response describes a motion or a direction, such as the W-shaped whisking of the matcha (Figure 4A), a concentric spiral pour (B), or perspective lines converging toward a painting's vanishing point (C), a 3D arrow traces it in the scene.
- **3D Object for Interactive Inspection.** A reference image externalizes absent content but shows it from a fixed viewpoint and scale [43]. When closer inspection from different viewpoints or scales is useful, we provide an interactive 3D object. Keeping the object visible throughout the narration could occlude the scene, so it remains hidden until the user chooses to open it. For example, pinching the reference image of the finished matcha in Figure 3C opens the 3D object shown in Figure 4D. The user can then rotate, resize, and position it anywhere in the scene to inspect the foam and texture more closely.
- **Panoramic View for Spatial Context.** A flat reference image shows what a place looks like but cannot convey its surrounding spatial context. We generate a panoramic view of the described environment and render it around the user, providing an immersive experience of the place (Figure 4E) [20, 52]. The panorama remains hidden until the user opens it by pinching the reference image (e.g., Figure 3D).
- **Keyword for Retention.** Visible and Absent content can also impose retention demands. A weight reading mentioned alongside a visible scale (Figure 4B) or the name and visiting details of a place shown in a reference image (F) can be difficult to retain as the narration continues. We supplement the base modality with keywords whenever such details appear.

4 VISTA WALKTHROUGH

This walkthrough shows how VISTA realizes situated visual modality adaptation. A user wearing an XR headset stands before Leonardo da Vinci's The Last Supper and asks, "Can you tell me about this?" The system answers in two run-time phases. Phase 1 converts the user's situated query into a navigable response overview (Figure 5). Phase 2 prepares and delivers the synchronized audio-visual presentation for each segment the user chooses to explore (Figure 6).

Phase 1: From Query to Overview. Once the system receives the egocentric image stream and speech query (❶), the Response Generator ❷ produces a detailed response grounded in the user's view. In parallel, Ack Generator ❸ immediately streams a short spoken acknowledgment ("Sure, let me look into it.") so the interaction remains conversational while the main pipeline runs. The Response Segmenter ❹ then decomposes the response into semantically coherent segments, assigns each a keyword and a response–scene-relation label, and generates an overview containing the segment count, the main target, the keywords, and a one-sentence gist. For instance, the *Artist and Location* segment is labeled *Absent* because it describes the painting's original setting, which is not present in the user's physical space. Then, the Object Detector ❺ localizes the painting for overview placement, while Text-to-Speech ❻ synthesizes the gist audio. Once both outputs are ready, the Sync Manager ❼ anchors the overview UI. It plays the spoken gist, allowing the user to inspect the response structure before selecting a segment to explore in detail.

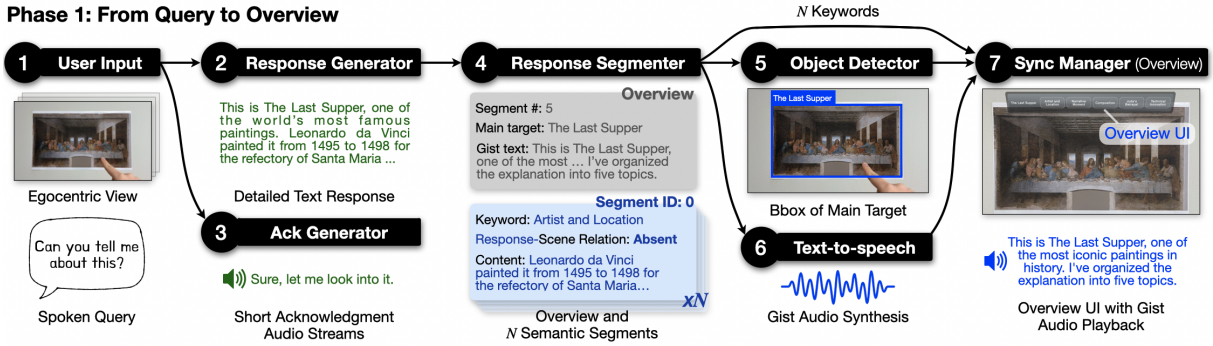


Figure 5: Phase 1 of the VISTA walkthrough. The pipeline turns a user query into a segmented response and presents a spatially anchored overview with gist audio.

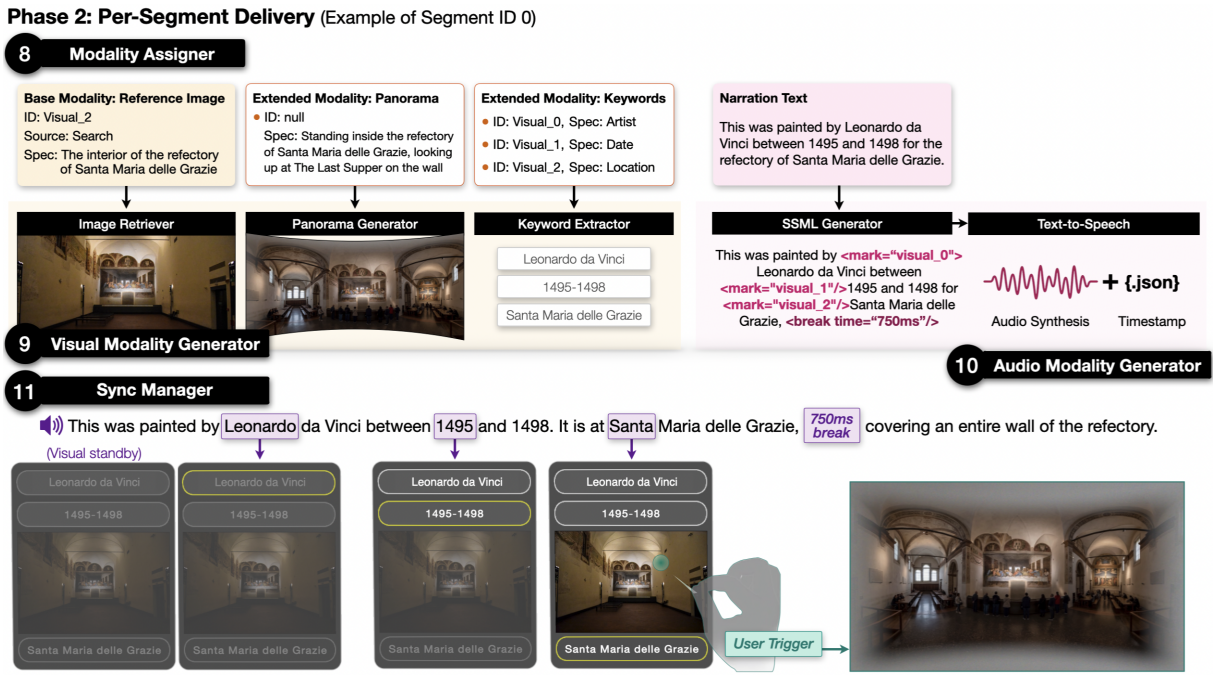


Figure 6: Phase 2 of the walkthrough. Using Segment 0 as an example, the pipeline assigns visual modalities, generates visual and audio assets in parallel, and synchronizes their delivery.

Phase 2: Per-Segment Delivery. As soon as segmentation finishes, all segments enter the Modality Assigner ❽ for background preparation. Here we trace Segment 0, which is labeled *Absent*. Following the mapping in Section 3.2, the Modality Assigner selects *reference image* as the base modality. It also adds *Panoramic View* as an extended modality for deeper spatial context and extracts three retention-critical *keywords*. It then rewrites the segment for natural spoken delivery.

Visual and audio assets are generated in parallel. The Visual Modality Generator ❾ routes each specification to the corresponding handler. For example, it fetches the refectory photograph, generates an immersive panoramic view, and extracts keywords. Meanwhile, the Audio Modality Generator ❿ inserts a timing mark at each point in the narration where a visual cue should be rendered in the XR view. It also adds an audio pause mark (e.g., 750 ms) to give the user time to process the visual information. Text-to-Speech then synthesizes the narration and returns the timestamp.

When the user selects the segment from the overview UI, the Sync Manager ⓫ begins playback. The three keywords and the reference image first appear in a semi-transparent standby state. As the narration progresses, each keyword activates at its first mention. When the narration reaches 'Santa,' the refectory image appears, and the narration briefly pauses. At the end of playback, the user can pinch the reference image to open the panoramic view. This interaction pattern enables VISTA to support immediate comprehension during the narrated explanation while reserving more visually disruptive expansions for deliberate user control.

5 IMPLEMENTATION

5.1 System Overview

We implement VISTA as a run-time response-delivery system for XR agents following the architecture shown in Figure 7. The XR client runs in Unity 6000.1.17f1 on an Android-based Samsung Galaxy XR headset.[1] It samples egocentric camera frames at 1 fps and sends them with the user's speech query to the server over WebSocket. The server executes the pipeline described in Section 4, including response generation, segmentation, modality assignment, and asset generation. Playback remains on the client, where the Sync Manager activates each visual cue when the narration reaches the corresponding content.

Two scheduling choices reduce idle waiting. (i) The client receives the overview and its spoken gist as soon as segmentation is complete. (ii) The user can inspect the response structure through the overview UI while the server prepares the segment assets. The server processes segments in parallel and generates their visual and audio assets concurrently. Each segment becomes playable when its own assets are ready.

To realize this delivery, we divide each response into segments that carry a single response–scene relation (Section 5.1.1), synchronize each visual cue with the moment its content is narrated (Section 5.1.2), and provide an overview through which the user explores the response at their own pace (Section 5.1.3). The supplementary material details the APIs, prompts, and output formats.

5.1.1 Response Segmentation

A free-form response must be segmented so that visual support can change with its response–scene relation. The structure a VLM already generates does not provide reliable boundaries. A single recipe step, for example, can describe cutting the visible vegetables, the target cut shape, and the exact amounts, spanning all three relations. Assigning one modality to such content leaves some demands unsupported, and showing several modalities together makes the user alternate among mental processes, adding switch costs [31, 80] and visual clutter. Segmenting every sentence avoids the mixing but fragments closely related information.

We set boundaries using transitions of the response–scene relation and breaks in semantic coherence. The segmenter processes the response one sentence at a time, identifying each sentence's main target and assigning its response–scene relation by the two distinctions in Section 3.1. It places a boundary whenever the relation changes or the semantic connection to the preceding sentence

[1] https://www.samsung.com/us/xr/galaxy-xr/

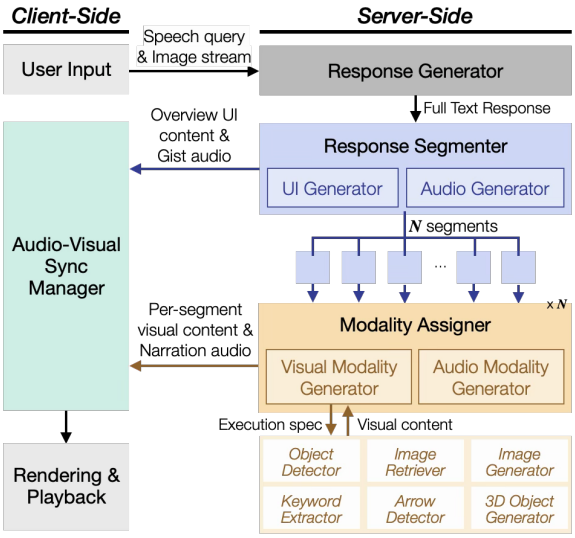


Figure 7: System overview of VISTA.

breaks. A step may thus be split when it mixes relations, and consecutive sentences that share a relation and remain coherent stay together. Each segment then maps to one base modality.

5.1.2 Audio-Visual Synchronization

The user must connect each visual cue to the corresponding part of the narration. Without temporal alignment, this matching becomes a demand in itself. We coordinate the two channels in time, exploiting the human capacity to process auditory and visual information in parallel [53, 55]. Two principles guide this coordination: (1) every visual cue is triggered at the first mention of its content, and (2) the narration's tempo adapts to the response–scene relation, since modalities differ in the time they take to process.

The Audio Modality Generator inserts a timing mark immediately before the first mention of each visual element. It also adds pauses based on the visual modality. For Visible content, spatial highlights and arrow lead-ins receive a 300 ms pause. This interval accommodates gaze shifts, as a saccade with fixation takes roughly 250 ms [64]. For Absent content, reference images receive a 750 ms pause for initial inspection [32, 64]. Keywords add no pause as they repeat the spoken content.

All cues first appear in a semi-transparent standby state. Each cue turns fully opaque as the narration reaches it, and returns to a de-emphasized state once the narration moves past. This incremental reveal limits visual overload and maintains temporal alignment between the two channels. Figure 8 illustrates this delivery for a Visible segment and Figure 6 ⓫ shows an Absent case.

5.1.3 Overview-Based Navigation

Continuous playback does not reveal the response structure or let the user jump to a specific segment. We present an overview with selectable segments so users can inspect a response's scope and choose where to begin. The layout stays consistent across responses of different lengths, topics, and structures.

When segmentation completes, the system presents the overview with a one-sentence spoken gist. It anchors the overview near the main target while preserving its visibility [78]. Each segment appears as a button labeled with a keyword. Selecting a button starts the segment's synchronized audio-visual delivery, and previously viewed segments remain available for replay.

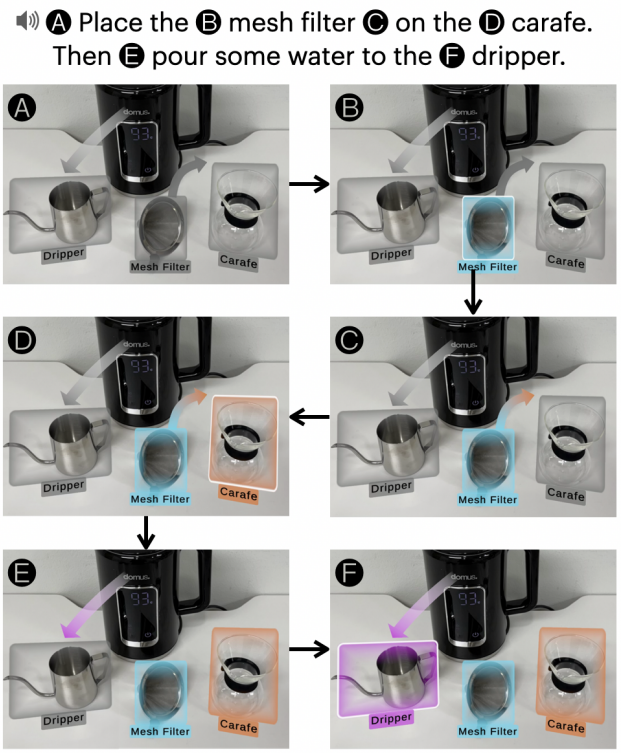


Figure 8: Audio-visual synchronization in a Visible segment. Timing marks (A–F) inserted in the narration activate each cue at the first mention of its content: cues start in a semi-transparent standby state and turn fully opaque when mentioned.

5.2 Pipeline Evaluation

We conducted a stage-level evaluation to examine how the pipeline performs across diverse situations and queries. Our goal was to characterize the automated response-delivery pipeline across a broader range of situations than the live user study could cover. We therefore evaluated each stage using fixed single-frame inputs, each consisting of one egocentric image and one query. Fixed inputs make the analysis reproducible and isolate errors in response segmentation, modality assignment, and visual and audio generation from errors introduced by head motion, depth sensing, or rendering. We collected 30 egocentric image–query pairs spanning six everyday categories, including cooking, art, shopping, and home repair. Two authors independently judged each output as correct or incorrect against a shared set of criteria, and another author adjudicated the items on which they disagreed. As each downstream stage depends on the correctness of the preceding one, we report conditional stage accuracy: each stage was evaluated only on segments that passed all earlier stages. Visual modalities are selectively invoked based on response content and modality requirements, making a single end-to-end accuracy difficult to interpret across different execution paths. Conditional evaluation instead characterizes the reliability of each stage when it is invoked. The supplementary material provides the full scenario set and judgment criteria.

Results. The 30 scenarios produced 161 segments (M=5.4, SD=1.2, range 4–8); Table 1 reports the accuracy of each stage. Segmentation failed when a single sentence mixed visible and absent content, such as one describing the jacket on the rack while also referring to how it drapes when worn.[2]

The Modality Assigner was almost always correct; accuracy

[2]All six segments that failed here were labeled absent. Following the conditional protocol, we exclude them from the Modality Assigner counts.

Table 1: System accuracy evaluated on 161 segments from 30 scenarios. Each stage is evaluated only on segments that passed all preceding stages. Rows shaded in light blue report the accuracy of the Modality Assigner ❽; rows shaded in light gray report the accuracy of the Visual Modality Generator ❾.

Stage	Correct / Total	Acc.
Response Segmenter ❹	155 / 161	96.3%
Modality Assigner ❽		
Visible	110 / 113	97.3%
Spatial highlight	94 / 110	85.5%
3D arrow	27 / 32	84.4%
Keywords	92 / 95	96.8%
Absent	30 / 30	100%
Reference Image	25 / 31	80.6%
⌞ Search	8 / 10	80.0%
⌞ Generated	17 / 21	81.0%
Panoramic view	8 / 10	80.0%
3D object	14 / 19	73.7%
Keywords	23 / 24	95.8%
Abstract	12 / 12	100%
Audio Modality Generator ❿	134 / 152	88.2%

dropped on the generation side. Failures for visible content were ones of coverage: a highlight marked only some of the objects a segment mentioned (e.g., four of six eggs), or an arrow's endpoint was detected in the wrong place. For absent content, the visual showed the right object or place but differed in detail: retrieved images placed the target in the wrong context (e.g., anemone species instead of how an anemone feeds), and generated images and 3D models missed the target state (e.g., a pouch sewn from a fabric other than the user's).

Audio generation for synchronization failed when the SSML markup ordered the cues incorrectly, omitted the pause before a reference image, or carried some timing marks that did not match the number of visual elements they were meant to trigger.

6 User Evaluation

We conducted a within-subjects study to evaluate whether VISTA improves comprehension and reduces perceived workload compared with an audio-with-text baseline. The study was approved by the university's Institutional Review Board.

6.1 Study Design

Baseline Design. We implemented an audio-with-text interface modeled after the built-in Gemini assistant of the Samsung Galaxy XR headset [27]. The agent narrates the full response, and a persistent panel provides playback controls (i.e., pause, resume, and replay) and an expandable text transcript. Both conditions share the same query input channels, response generation model, prompt, and TTS engine. They differed only in response delivery. The baseline delivered the response as one continuous narration. VISTA divided the generated response into segments by response–scene relation and synchronized visual cues with the narration. This comparison isolates the combined effect of VISTA's response-delivery design.

Condition Comparability. Figure 9 shows the delivery timeline measured over the 24 study queries per condition. Acknowledgment onset (Baseline: 4.97 vs. VISTA: 4.67 s) and main audio onset (16.90 vs. 16.64 s) did not differ between conditions (both $p > .14$). The main audio was also comparable in total duration (97.3 vs. 98.7 s) and word count (227 vs. 211 words; both $p > .10$). In VISTA, segment preparation overlapped with the acknowledgment, gist, and overview exploration: on average, the first segment became playable at 30.7 s, before participants' first selection at 38.0 s. This rules out response latency and the amount of narrated content as explanations for the differences we report below.

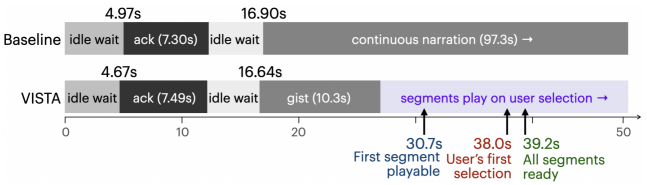


Figure 9: Response delivery timeline, averaged over the 24 user-study queries per condition. The x-axis is time after the end of the user's query.

Procedure and Tasks. After a brief tutorial on the headset's hand tracking and voice interaction, participants completed one declarative task and one procedural task in each condition, for four tasks in total (Figure 10). Both task types elicited long, information-dense responses grounded in the participant's physical surroundings. In the declarative task, participants stood before a printed reproduction of Leonardo da Vinci's The Last Supper or the Korean royal screen painting Irworobongdo and asked the agent to explain it. In the procedural task, they followed the agent's instructions to prepare pour-over coffee or matcha tea. We organized the tasks into two matched sets, and each condition used a different set. We counterbalanced condition order and set to reduce content-specific learning effects. Participants completed a comprehension quiz and questionnaires after each task. After all four tasks, we conducted a 15-minute semi-structured interview. Each session lasted approximately 90 minutes.

Participants. We recruited 12 participants (4 female, 8 male; mean age = 24.58, SD = 3.00) from our institution via word of mouth. All participants had prior XR experience. Each participant was compensated US$20 for their time.

Measurements. We measured objective comprehension with a 10-item, four-choice *comprehension quiz* administered after each task [42]. Responses were generated at run time, so the system generated each quiz from the response received by that participant. All quizzes used the same prompt, item count, and format (see supplementary material). We instructed participants to leave items blank when unsure. To assess subjective information acquisition, we measured *perceived knowledge gain* with three cognitive learning outcome items [66] and *perceived information quality* through ratings of relevance, detail, and clarity [1, 22].

We measured perceived workload with NASA-TLX [30] as an indicator of cognitive load. We also assessed overall usability with SUS [4, 12]. The semi-structured interviews examined per-task impressions and cross-condition comparisons.

Data Collection and Analysis. After each task, participants com-

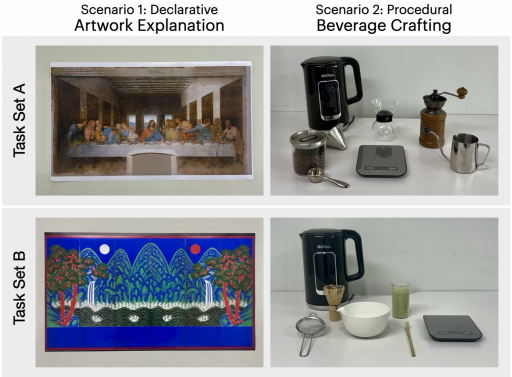


Figure 10: User evaluation tasks. Each participant completed all four tasks. Task sets A and B contain matched content, and each condition used a different set.

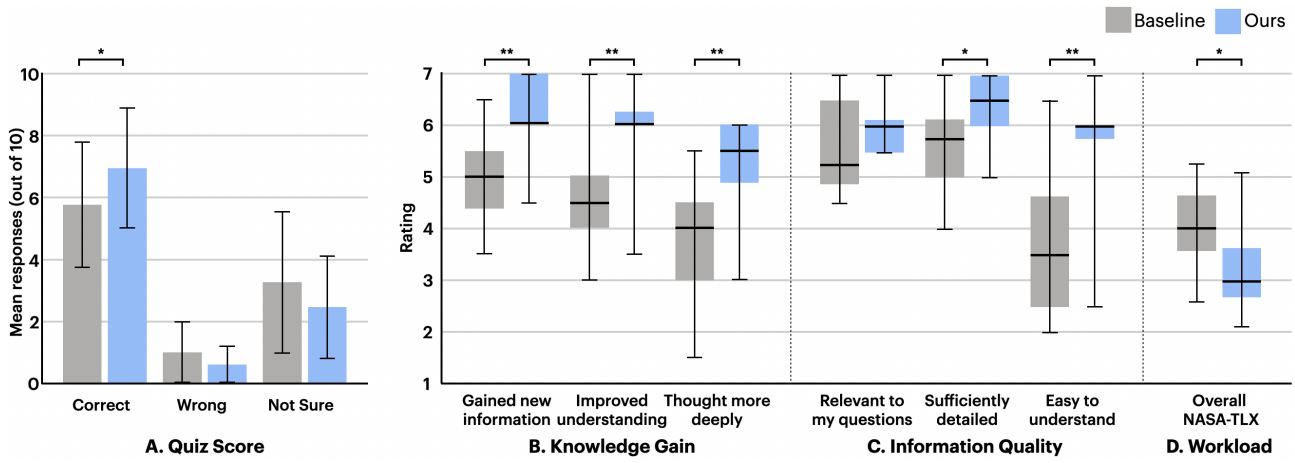


Figure 11: Quantitative results. (A) Comprehension quiz responses. (B) Perceived knowledge gain. (C) Perceived information quality. (D) Overall NASA-TLX workload. Bars in A show means with standard deviations. Boxes in B, C, and D show medians and interquartile ranges. Higher values indicate better in A, B, and C. Lower values indicate lower workload in D. Brackets mark significant differences between conditions based on Holm-adjusted p-values (* $p < .05$, ** $p < .01$, *** $p < .001$).

pleted a post-task questionnaire. Unless otherwise noted, questionnaire items used 7-point Likert scales. We compared conditions using Wilcoxon signed-rank tests and reported Holm-adjusted p-values. Quiz accuracy and overall NASA-TLX workload were the primary outcomes, representing the study's comprehension and cognitive-load effects. All remaining measures were exploratory. We applied Holm–Bonferroni correction across the two primary outcomes and within each exploratory scale family. Full test statistics and matched-pairs rank-biserial correlations (r) are provided in the supplementary material. We analyzed the interview data using reflexive thematic analysis [9]. One author coded the transcripts inductively, and the authors refined the themes through discussion.

6.2 Results

Participants achieved higher comprehension scores and reported lower perceived workload with VISTA than with the baseline. We first report results from the comprehension quiz and questionnaires, followed by interview findings that describe how participants experienced the two response-delivery conditions.

6.2.1 Quantitative Results

Figure 11 shows that participants using VISTA achieved higher comprehension scores and reported greater knowledge gain with lower perceived workload than in the baseline condition.

Response Comprehension. Participants answered significantly more quiz questions correctly with VISTA ($M = 6.92$, $SD = 1.93$) than with the baseline ($M = 5.75$, $SD = 2.01$; $p = .048$, $r = .67$). They also rated all three perceived knowledge gain items higher with VISTA (all $p \leq .008$). Detail ($p = .020$) and clarity ratings ($p = .009$) were higher with VISTA. As expected, relevance ratings did not differ between conditions ($p = .125$), since both conditions used matched tasks and the same response-generation setup.

Perceived Workload. Participants reported lower overall NASA-TLX workload with VISTA ($M = 3.14$, $SD = 0.83$) than with the baseline ($M = 4.04$, $SD = 0.79$; $p = .032$, $r = -.77$). Mental demand ($p = .006$), temporal demand ($p = .021$), and frustration ($p = .020$) were lower with VISTA. Participants also rated their task performance higher ($p = .006$).

Usability. The mean SUS scores were 74.9 ($SD = 14.7$) for VISTA and 69.7 ($SD = 16.9$) for the baseline, and both exceeded the acceptability threshold of 68 [5].

6.2.2 Qualitative Findings

Semi-structured interviews revealed how different aspects of VISTA shaped participants' experiences of receiving XR agent responses.

Information in each response–scene relation was easier to follow. In the baseline, participants handled mental mapping, imagery, and retention on their own while the narration continued. VISTA resolved the demand each response–scene relation imposed and improved comprehension. We describe what participants experienced in each relation below.

- **Visible.** Participants located visible targets without searching the scene on their own (10/12). In the baseline, participants searched while the narration continued and sometimes missed what followed (3/12). Since VISTA spatially highlighted each target as it was mentioned, P7 could put that effort into understanding the response. P4 described this experience as *"joint attention with the agent"* and compared it to *"following a kind guide."* Similarly, in the baseline, participants inferred the intended motion and the direction to look from the narration alone. In our system, 3D arrows showed both directly and helped them interpret the narration more accurately (5/12).
- **Absent.** Our system externalized absent content so participants could see and understand what the narration described (8/12). In the baseline, participants looked around to check whether the described content was in the scene when the response shifted to absent content (5/12). They sometimes lost track of the narration or returned to the transcript (3/12). With VISTA, participants could inspect absent content in reference images while the narration continued. P2 could linger over information that *"would have slipped past as audio."* All participants also explored the panoramas and 3D objects. P8 said that manipulating 3D objects helped them understand unfamiliar objects and tasks more intuitively, and P3 described the panoramas as immersive and said they made the explanation more memorable.
- **Abstract.** Participants retained unfamiliar terms, proper nouns, and quantities throughout the response while their attention stayed on the physical scene (10/12). In the baseline, participants often missed these details in the audio, and their attention repeatedly shifted from the physical scene to the transcript (2/12). With ours, anchored keywords made complex names and numbers easier for P2 to follow and helped P5 recall the content at a glance after the narration ended.

Taken together, VISTA addressed the comprehension demand associated with each response–scene relation and adapted its support as

the response moved from one relation to another. In the baseline, participants had to determine how to follow the response as its demands changed, which distracted them from the narration (5/12). VISTA made this judgment for them by identifying each segment's dominant demand and adapting the presentation accordingly. This reduced the effort of shifting among mental mapping, imagery, and retention (6/12), a process that prior work has shown to impose measurable cognitive costs [31, 80]. Participants could focus on the narration and develop a deeper understanding of the agent's response. They described the experience as offloading the effort of acquiring information onto the agent (P7), work they had to *"figure out myself"* in the baseline (P4).

Temporal synchronization reduced attentional overhead and increased trust. Participants explicitly appreciated that each visual cue appeared when the corresponding content was narrated (10/12). Some said this increased trust in the agent (4/12), while others said it freed attention for deeper understanding (3/12). P1 described it as *"like an expert standing right beside me, pointing to each thing at exactly the right moment."* P6 added that synchronized cues reduced the need to manually connect spoken information to the physical world and left them *"more confident answering the quiz."*

The overview UI gave users more control over pacing. Participants valued the overview UI as they could (i) scan the response topics before playback (9/12) and (ii) choose a starting point when listening again (7/12). Specifically, participants used the overview as a progress tracker in procedural tasks (*"I can see the five steps. Just do these and I'm done!"*, P10). This suggests that the overview helped users traverse the explanation more actively rather than receiving it as a single forced stream.

7 Discussion

VISTA improved comprehension and reduced workload as an integrated system. Our findings suggest that an agent answering open-ended questions must coordinate its delivery at run time by dividing each response, choosing a visual form for each part, and timing each cue as the narration unfolds. The following subsections discuss future directions raised by participant feedback and limitations of our approach.

7.1 Toward User-Aware Response Delivery

Participants differed in where comprehension broke down. P6 retained numerical details well, so the keywords added little for them. Spatial highlights helped them locate referents among unfamiliar tools. P1 struggled to picture spaces from speech, and panoramas clarified the scale and layout of those spaces. They found 3D objects less valuable. These patterns are consistent with prior findings that visual search, working-memory capacity, and mental imagery ability differ across individuals [7, 19, 44, 79].

Our approach selects support from the response and its relation to the physical scene. This decision rule provided a useful starting point across the individual bottlenecks observed in our study without requiring a user model. The interviews also revealed needs that response content alone could not indicate. P1, who was unfamiliar with art terminology, found the overview keywords too detailed to choose a starting segment. Future systems could refine the initial selection using user-side signals such as prior knowledge, interaction history, and the desired amount of explanation. These signals could adjust keyword detail, access to extended modalities, and the overview structure.

7.2 Balancing Automation and User Control

VISTA automated the selection of visual support and its synchronization with the narration, reducing the user's comprehension burden. Participants controlled pacing, order, and level of detail through the overview and optional visuals (i.e., panoramic views and 3D objects). This division suggests a practical starting point for automation in XR agents. Since VISTA cannot know how quickly a user comprehends an explanation or how much detail they need, users should remain able to adjust these aspects.

Future work could lower the cost of exercising this control. Adjusting the delivery currently requires manual actions, such as opening the overview or pinching a reference image. Participants who rarely used XR (P2, P8) or who felt more comfortable speaking found these manipulations effortful (P5). Lighter mechanisms, such as short voice commands for tempo (e.g., "next" to play the following segment) and depth (e.g., "tell me more about that"), could make these adjustments more immediate and intuitive. Complementary signals from XR sensing offer another strategy. Gaze dwelling on a highlight, repeated fixations on a keyword, or a shift away from the target region could suggest that a segment needs more time or less detail. The system could act on these signals provisionally, and the user would retain the final decision.

7.3 Calibrating Trust in Visual Support

Participants used spatial and temporal alignment to judge the agent's reliability. P1 reported that highlights landing on the correct object confirmed that their understanding matched the agent's intent, and four participants associated synchronized cues with greater trust. P3 also imagined using reference images to verify ingredient identity in future cooking scenarios. This benefit, however, carries a risk. In our technical evaluation, reference image generation reached 80.6% accuracy and 3D arrows 84.4%. Though these accuracies reflect the current state of the underlying generative models, an inaccurate cue could reinforce a wrong belief if users read visual support as implicit confirmation. P7 raised this concern, noting that misplaced trust during furniture assembly could lead to irreversible mistakes. Future systems should help users calibrate trust in visual support through confidence indicators, source attribution, or controls to correct or dismiss inaccurate cues.

7.4 Granularity of Response–Scene Relations

Our three response–scene-relation types provide an initial representation of how response content relates to the physical scene. This granularity offered a useful basis for selecting support in our study. Future systems could model these relations more precisely. One direction is to decompose the Absent type. Absent content may describe an entity located elsewhere, a past or future state of a visible object, or another instance of a familiar category. These cases differ in their spatial, temporal, and instance-level relations to the scene. Future systems could model each dimension separately. Along the temporal dimension, a future state of the user's bowl could be overlaid directly on the bowl, an object could be animated to show how its state changes over time, and a slider could let users inspect its prior states. These representations could connect absent states more closely to the physical objects they describe.

8 Conclusion

We present VISTA, an XR agent system that delivers free-form, information-rich VLM responses as synchronized audio-visual presentations. Our method segments each response based on the response–scene relation of its content and adaptively generates matched visual support at runtime. We also synchronize each visual cue with the moment its content is narrated and provide an overview that lets users navigate the response at their own pace. We verify VISTA 's effectiveness by evaluating a fully working prototype in a user study, showing that it provides higher comprehension and lower perceived workload than an audio-with-text baseline.

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